How Much Late-Game Shooting Carries Over? Shot Volume, Accuracy and Free Throws in NBA Play-by-Play, 2012–26

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Abstract

Some players take more of their team's shots late in close games than their role predicts. Is that a stable tendency, and is late-game accuracy one? Using win-probability leverage splits for 2012-13 to 2025-26 and play-by-play for 13,508 games, the study measures each player's late-game shot volume against what his role predicts and asks how well it carries over to the next season. Late-game shot volume carries over modestly, with r = 0.29 (95% interval 0.22 to 0.35) on 1,237 season pairs from 374 players, far above the chance range and well below the 0.85 of ordinary shot volume on the same pairs. Late-game accuracy shows no detectable carryover (shot-making r = −0.005, free throws under pressure r = −0.02), but at these attempt counts even a real 3-point spread between players would show as an r of at most about 0.10. League-wide, free-throw percentage in high-leverage moments runs 1.20 points below each shooter's own normal rate (95% interval −1.69 to −0.71), and no gain is detected where pressure is low. In a side study, combine shooting drills and college three-point percentage both carry information about NBA three-point percentage, and the data cannot rank them. The results are descriptive. The original pilots' thresholds were set after seeing the data and are retired.

1. Question

The project began from a scouting puzzle. Players can look sharper in workouts and practice than in games, and the question was whether a difference by game situation can be pinned to individual players. It narrows to two measurable questions. Is a player's late-game shot volume, beyond what his role predicts, a tendency that carries over from one season to the next? And does late-game accuracy carry over? A league-wide comparison of free-throw percentage across settings, from preseason to the playoffs, and a side study of combine shooting drills complete it. What is measured is a late-game shot-attempt tendency.

2. Data

Input	Source	Coverage
Player totals by win-probability leverage	pbpstats (api.pbpstats.com), 112 responses	7,162 player-seasons in the regular-season and playoff lists, 2012-13 to 2025-26
Play-by-play, win probability, box scores	ESPN game summaries, copied without unused third-party fields	13,508 games with play-by-play; 2,988,941† shots, 612,339† of them free throws
Combine shooting drills	NBA Draft Combine tables (public mirror of stats.nba.com)	526 entries from 496 players; 202 entries (193 players) in the regression
College shooting	Bart Torvik college player seasons, 2008–2026	college 3P% and FT% of combine participants

Gaps are disclosed rather than filled. ESPN has no usable win probability for 1,481 games, which leave the garbage-time and normal rows. Four of 85 playoff games in 2025-26 have no ESPN file. pbpstats returns at most 500 players per split, and 21 of 112 files reach that cap. 14 games whose play-by-play falls 5 or more points short of the final score are flagged and kept. Every input's sha256 is recorded in the repository's data manifest.

3. Methods

3.1 Late-game shot volume and its carryover

Late-game means possessions in pbpstats' high and very-high win-probability leverage buckets, and medium leverage is the ordinary game. For each player-season the measure is log(late-game rate ÷ medium-leverage rate), with rates per 100 team possessions on court so that playing time drops out, and half an attempt added to the late count. Role is removed by regressing on the same season's medium-leverage usage, shot mix, assists, free-throw rate and percentage, minutes and season. A player needs at least 120 late-game and 1,000 medium-leverage team possessions on court in both seasons of a pair, a floor on exposure rather than on shots. Shot attempts are the headline volume measure, with usage, usage without free throws and jump shots reported beside them.

Carryover is the Pearson correlation of the role-adjusted measure in season t with season t+1 over player-season pairs. Intervals come from a bootstrap that resamples players with all their pairs (5,000 draws). The chance range is ±1.96 SD of r over 2,000 shuffles that move each player's next-season values, as a block, to another player with the same number of pairs. The benchmark is ordinary shot volume (medium-leverage attempts per 100 team possessions) on the same pairs.

3.2 Late-game accuracy

Shot-making is effective field-goal percentage minus pbpstats' expected value, late minus medium leverage (at least 40 late attempts). Free throws under pressure are late minus medium free-throw

percentage (at least 15 late and 100 medium attempts). Their carryover is measured the same way.

3.3 Free throws by setting

Each free throw is compared with the shooter's own normal rate, which is his medium-leverage percentage that season (pbpstats rows) or his career percentage in normal game states (ESPN rows). The estimate is attempt-weighted, so it leans on the players fouled most in a setting. The settings are preseason, garbage time (win probability of 95% or more), the empty arenas of 2020-21, high and very-high leverage, NBA clutch time (the last 5:00 of the fourth quarter or overtime with the score within 5) and the playoffs. Intervals resample players.

3.4 Combine side study

A weighted least-squares regression of NBA career three-point percentage, weighted by NBA attempts, on combine drill percentage, college three-point percentage and college free-throw percentage, each standardized. The sample is 202 combine entries from 193 players with at least 20 drill shots, 50 college free-throw and three-point attempts, and 100 NBA three-point attempts after the combine. It is a survivor sample. Intervals are clustered by player.

3.5 Evidence type

Every result is descriptive. The pilots' pass bars (0.15 and 0.30), the exposure floors and the choice of shot attempts as the headline were set after the pilot data and are disclosed as such, and the bars are retired. The checks added in the revision were declared before they ran, with the declaration's sha256 recorded first. The specification registry lists 315 specifications.

4. Results

4.1 Late-game shot volume carries over modestly

Measure (role-adjusted, season t vs t+1)	r	95% interval	Pairs · players
Shot attempts (headline)	0.29	0.22 to 0.35	1,237 · 374
Usage (shots, free throws, turnovers)	0.32	0.25 to 0.37	1,237 · 374
Usage without free throws	0.27	0.21 to 0.33	1,237 · 374
Jump shots	0.27	0.19 to 0.34	1,237 · 374
Shot-making vs expected eFG	−0.005	−0.11 to 0.09	351 · 122
Free-throw % under pressure	−0.02	−0.12 to 0.07	387 · 129
Benchmark: ordinary shot volume, same pairs	0.85	0.83 to 0.87	1,237 · 374

Table 1. Season-to-season correlation of each role-adjusted measure. Intervals resample players with all their pairs; the chance range for shot attempts is about ±0.06 (shuffle SD 0.029).

Late-game shot volume carries over modestly, with r = 0.29 for shot attempts, far above the chance range and well below the 0.85 of ordinary shot volume on the same pairs. In basketball terms, one SD of the persistent tendency is ×1.13 up or ×0.89 down against role (+13% / −11%), about 3.9 more or 3.4 fewer late shots a season for the median qualifying player-season. A player 10% above his role-expected late volume this season is expected to be about 2.9% above it next season (slope 0.30). Usage counts free throws, and many late free throws come from intentional fouls, so it drops from 0.32 to 0.27 without them. Players who changed teams carry it over about as much (0.28, with stayers minus movers +0.02, 95% interval −0.11 to +0.15). Richer role models leave it in place, at 0.286 with team-hierarchy controls and 0.311 with an out-of-fold boosted-tree role model.

4.2 Late-game accuracy shows no detectable carryover

Shot-making over expected (r = −0.005, 95% interval −0.11 to 0.09) and free throws under pressure (r = −0.02, −0.12 to 0.07) show no detectable carryover. These intervals bound real differences rather than rule them out. Late-game attempts are few. At these counts, a true 3-point spread between players would show as an r of at most about 0.12 for shot-making and 0.10 for free throws. A 75% shooter on 15 free throws has an SD of about 11.2 points in observed percentage, so one season of pressure free throws says little about a player.

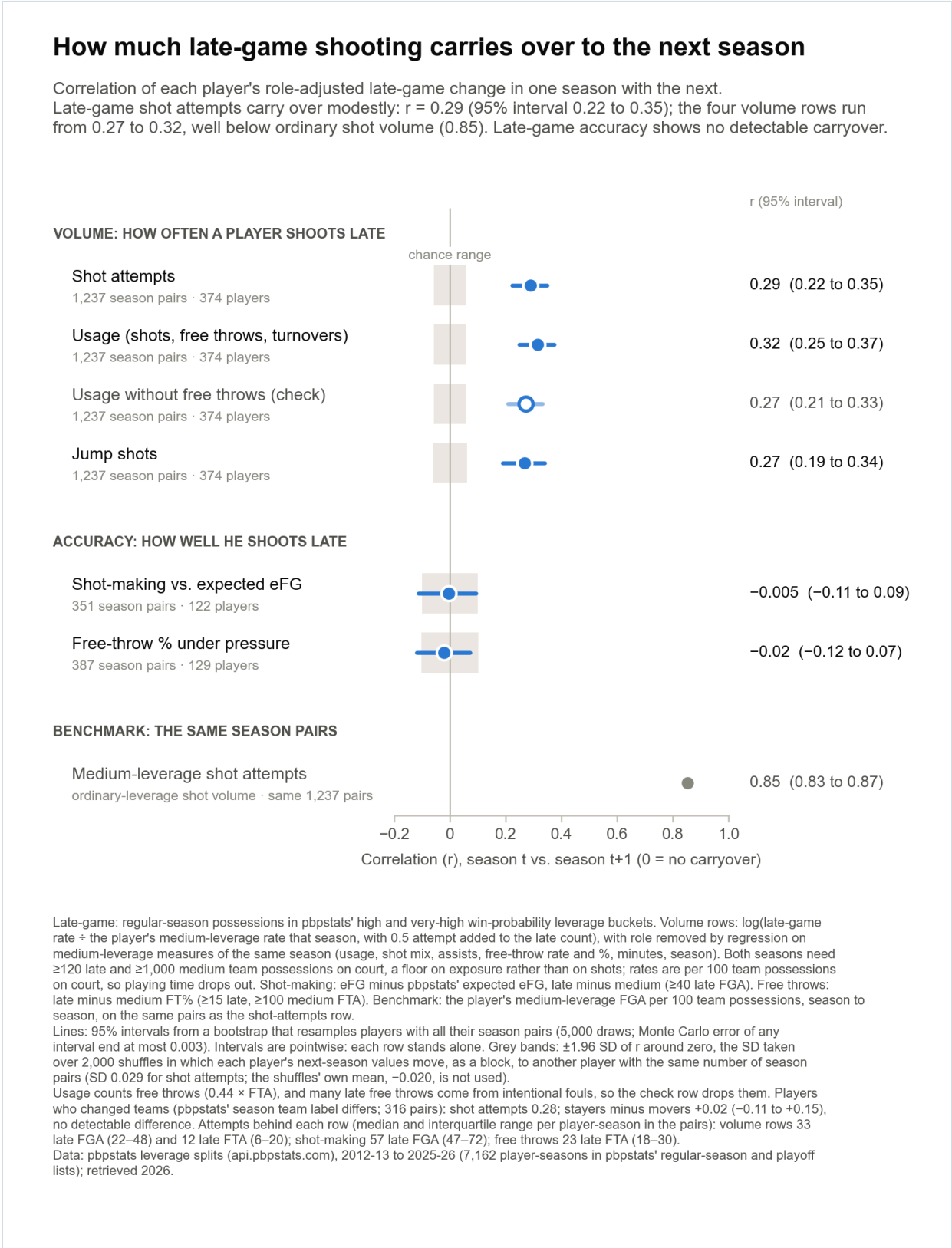


Figure 1. How much late-game shooting carries over to the next season. Lines show 95% intervals from a player-resampling bootstrap, and grey bands show ±1.96 SD of r under a player-block shuffle, around zero. The grey bottom row is ordinary shot volume on the same pairs.

4.3 Free throws by setting, against each shooter's own normal rate

Setting	Change in FT% (points)	95% interval	Attempts
Preseason	−1.14†	−1.79 to −0.49	20,871
Garbage time (win probability ≥ 95%)	−0.10†	−0.37 to +0.17	110,163
Empty arenas, 2020-21	−0.23	−0.71 to +0.27	13,074
High leverage	−1.20	−1.69 to −0.71	27,882
Very high leverage	−2.98	−4.19 to −1.85	6,387
NBA clutch time	−1.48†	−2.02 to −0.95	33,375
Playoffs	−1.08	−1.53 to −0.62	50,590

Table 2. Attempt-weighted change in free-throw percentage against each shooter's own normal rate; intervals resample players. Player-weighted versions: high −0.97, very high −3.02, clutch −1.39.

League-wide, free-throw percentage runs below each shooter's own normal rate in high-leverage moments (−1.20 points), more so in very-high-leverage moments (−2.98; very high minus high −1.79, 95% interval −3.05 to −0.51), in clutch time (−1.48) and in the playoffs (−1.08). No gain is detected where there is less pressure (garbage time −0.10, the empty arenas of 2020-21 −0.23). Preseason is lower too (−1.14). Rust could explain it, but that is untested. At league volume, the high-leverage dip is about −0.85 points per team-season.

The dip is not separated from timing. Late free throws come largely from trailing teams fouling, so who gets fouled, the type of trip and the time of game travel with pressure. Against each player's normal fourth-quarter rate, the clutch dip is −0.47 (95% interval −1.20 to +0.27). The clutch rule also has a quirk at the limits of its 5-point window, where whether a later free throw of a trip counts as clutch depends on the earlier ones. A rule that reads only the score before the trip gives −1.64 (−2.19 to −1.11) instead of −1.48, so the shown rule likely understates the dip a little (about −1.5, or −1.6 under the outcome-free rule).

4.4 Side study: combine drills and college shooting

Among the 193 players who reached 100 NBA three-point attempts, combine drill accuracy carries information beyond college three-point shooting: +0.83 NBA three-point percentage points per SD (95% interval +0.30 to +1.41), next to +0.69 for college three-point percentage (+0.10 to +1.24) and +0.18 for college free-throw percentage (−0.34 to +0.74). The two lead inputs cannot be ranked, since drill minus college three-point percentage is +0.14 (−0.66 to +1.06). Each coefficient is a partial association, not an effect of the drill, and the inputs overlap only modestly (VIF about 1.3). The sample is 57% guards, against 44% of all drill participants.

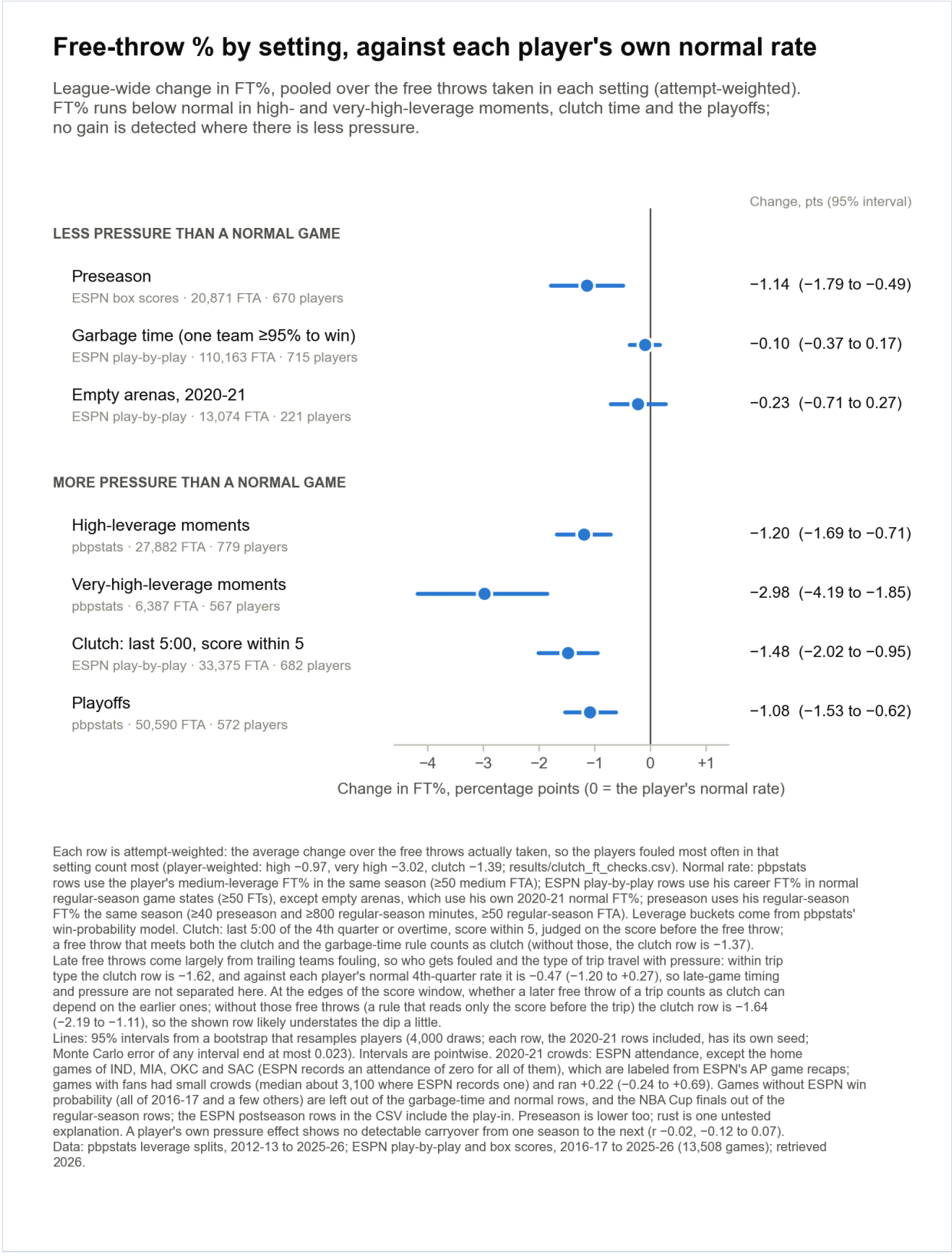


Figure 2. Free-throw percentage by setting, against each player's own normal rate, attempt-weighted, with 95% intervals from a player-resampling bootstrap and attempt counts. The clutch-time dip is not separated from late-game timing.

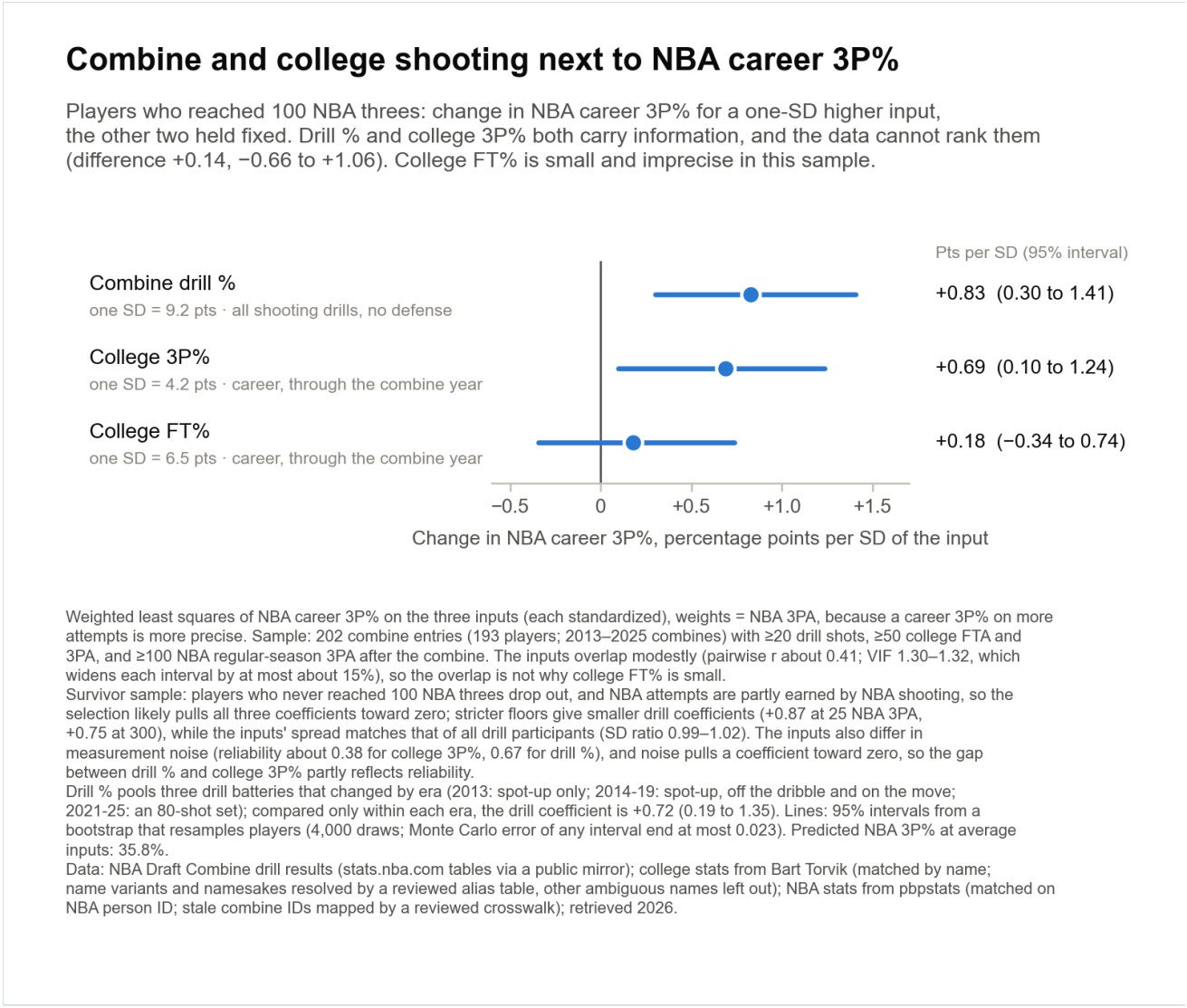


Figure 3. Combine and college shooting next to NBA career three-point percentage. Weighted least squares (weights = NBA attempts) with standardized inputs and player-clustered 95% intervals, on 202 entries from 193 players (a survivor sample).

5. Checks and diagnostics

- **Official data as the check.** Aggregated play-by-play matches the box score exactly for 98.9%† of player-games' three-point attempts and 99.3%† of their field-goal attempts, over 272,213 player-games compared.
- **Second route.** A cross-check recomputes the headline numbers a second way, and all 31 agree.
- **Placebo.** A role-model placebo over 15 seeds runs −0.02 to +0.05 for late shot attempts, inside the chance range.
- **Exposure.** The carryover rises with exposure (0.20 at a floor of 60 late possessions, 0.29 at 120, 0.38 at 240), and persistence is weaker over longer gaps (one-way ICC over all qualifying seasons 0.18, 95% interval 0.13 to 0.23).

- **Rebuild.** One command rebuilds every table, figure and document from empty results, and two consecutive rebuilds give byte-identical results. 848 tests pass, and the share scan, vocabulary lint and claim lint report zero hits.

6. Data corrections

A data validation on October 21–22, 2026, with a second checker that re-derived each issue with its own code, found the fixes below. Each was applied as one step of a fix ledger whose step 0 reproduces the earlier tables cell for cell. The largest was in ESPN's play-by-play, which stored zero points for made free throws from 2016-17 to November 2019, so 152,257 made free throws carried the score after the shot instead of before it, which moved free throws in and out of clutch time and garbage time. Crowd labels for four teams' 2020-21 home games now come from game recaps instead of a recorded attendance of zero. The combine join gained an ID crosswalk and a college alias table. Hygiene steps removed Cup finals from regular-season contexts, games without win probability from the garbage-time and normal rows, end-of-quarter heaves from field-goal attempts, and duplicate rows the box score confirmed.

Quantity	Before	After
Clutch-time free throws (change in FT%)	−1.84 (33,104 FTA)	−1.48 (33,375 FTA)
Garbage-time free throws	+0.00	−0.10
Empty arenas, 2020-21	−0.22	−0.23
Combine regression: players; drill coefficient	186; +0.88	193; +0.83
Combine regression: college 3P% coefficient	+0.73	+0.69
Shots in the play-by-play	2,988,953	2,988,941

Table 3. Unchanged by every data fix: the shot-attempt carryover (r = 0.29, 0.22 to 0.35), the accuracy rows, the leverage free-throw rows and the game count.

Several earlier claims are retired:

- that the volume result "passed" a bar or was pre-registered (the bars were set after the pilots)
- "7 to 11 standard deviations above chance" (it divided by an unclustered shuffle SD)
- "about 12 to 15% more or fewer shots per SD" (now +13% / −11%, computed in code)
- a narrower placebo range, and a count of 13,590 games (13,590 cached files, 13,508 games with play-by-play)

7. Limitations

Limit	Mechanism	Likely direction
Role the model misses	Closing-lineup spots, play calls and putbacks for bigs persist across seasons.	Overstates r. Team changers, team-hierarchy controls and a boosted role model narrow it but cannot rule it out.
Who gets fouled and when	Late free throws come largely from trailing teams fouling, so pressure travels with the shooter, the trip type and the timing.	The pressure reading likely overstates the part that is pressure rather than timing or fatigue.
Leverage is one model's	pbpstats' win-probability buckets define late-game; there is no second source.	Unknown.
Noisy late samples	Few late attempts per player-season.	Pull the accuracy rows toward zero, so any real carryover in accuracy is likely understated.
Exposure floors	The floors keep rotation players, and r rises with exposure.	For all players, bench included, r would likely be lower than the headline.
Bundled settings	The 2020-21 crowd and preseason comparisons also differ in schedule, health protocols, lineups and effort.	Unknown.
Combine selection	The sample requires 100 NBA threes, which NBA shooting partly earns; the drill batteries changed by era.	All three coefficients are likely pulled toward zero (stricter floors give smaller drill coefficients).

Intervals reflect sampling noise only, not who is in the sample.

8. How a team could use it

- **Late-shot roles (predictive).** A player's late-game shot share is a modestly stable tendency, so pool several seasons before reading anything into one.
- **Free throws (predictive).** Expect a small league-wide dip in high-leverage moments, and do not label individual players from their makes.
- **Draft (predictive).** For players who will take NBA threes, combine drill percentage is an input next to college three-point percentage. This sample cannot rank the two, and it is mostly guards.
- **Causal claims.** None, because the free-throw rows do not separate pressure from timing.

9. Reproducibility and how this was built

One command, `make rebuild`, empties the results and figures folders and rebuilds every table, figure and document from a private data root. Every random step is seeded, and two consecutive

rebuilds give byte-identical results. Every number, season and floor in a figure's text is built from a results table or a code constant, and the build stops on a typed digit. Raw caches, player-level tables and anything that names a player stay in the private data root.

The project started from the author's question about why players look sharper in workouts than in games, and the October revision followed his instructions: apply every applicable lesson from his class lectures, add no new research questions, and keep unused third-party data fields and player names out of the shared copy. AI coding tools (Claude Code) wrote the code and drafted this paper from the project's results files, and AI review agents, with a second checker, ran the validation and verification passes. The revision applied lessons from Syracuse coursework in statistics, sport analytics, information studies and communication: intervals that resample players, chance ranges from block shuffles, nulls stated as bounds, retired claims named with how each was found, and figure text built from data.

† Moved by a later data fix and not yet reproduced by an independent second checker. The free-throw intervals and the role-model correlations also include a labeled method step whose expected values in the test suite await the author's sign-off.

Data sources (retrieval dates)

- pbpstats player totals by leverage (api.pbpstats.com), 2012-13 to 2025-26: October 13, 2026.
- ESPN game summaries (play-by-play, win probability, box score): pulled October 13, 2026; copied without unused third-party fields October 22, 2026.
- NBA Draft Combine shooting drills, stats.nba.com tables via a public GitHub mirror: October 13, 2026.
- Bart Torvik college player seasons, 2008–2026: built August 10, 2026. Basketball-Reference player totals (data dump dated April 13, 2026), for reconciliation counts.

Appendix

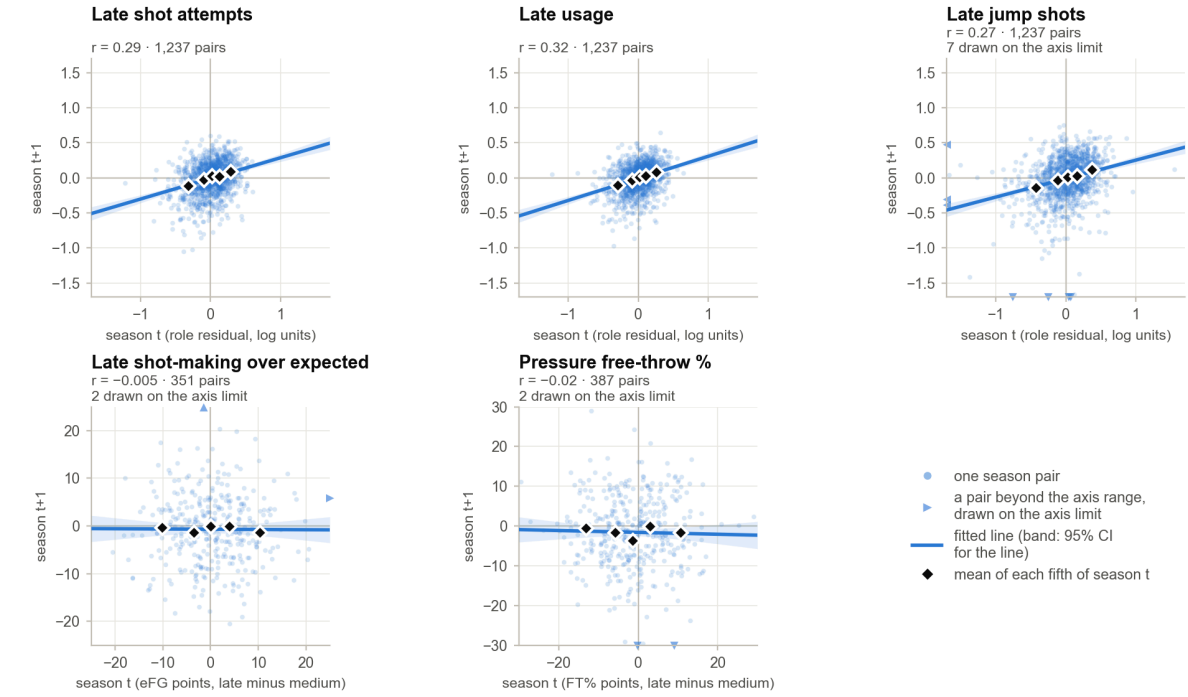


Figure A1. Every season pair behind Figure 1, with a fitted line, its 95% band and the mean of each fifth of season t. The band is narrow because there are many pairs, not because the relationship is strong: for late shot attempts about 8% of next season's variation (5% to 12%) is shared with this season's. No player identifiers are plotted.